Document serial number: 000010343062

HeatWave Library Reference

000010343062

Document serial number: 000010343062

© 2016 Sony Interactive Entertainment Inc.
All Rights Reserved.
SIE Confidential

Document serial number: 000010343062

Table of Contents

Data Types **5**
- float32_t 6
- SceHwError 7
- SceHwFileFunctions 8
- SceHwHandle 9
- SceHwInfo 10
- SceHwInitParams 11
- SceHwMemoryFileInfo 12
- SceHwMemoryFunctions 13
- SceHwSfx 14
- SceHwSfxCallbackInfo 15
- SceHwSfxCallbackType 16
- SceHwSfxInfo 17
- SceHwSfxPlayParam 18
- SceHwSfxState 19
- SceHwSound 20
- SceHwSoundInfo 21
- SceHwSource 22
- SceHwSourceInfo 23
- SceHwSourceRuntimeStreamCallbackInfo 24
- SceHwSourceRuntimeStreamParam 25
- SceHwSourceType 26
- SceHwSystemCallbackInfo 27
- SceHwSystemCallbackType 28
- SceHwSystemOutputType 29
- SceHwVector 30
- ScewHwSourceRuntimeStreamCallbackType 31

Callback Functions **32**
- SceHwFileClose 33
- SceHwFileOpen 34
- SceHwFileRead 35
- SceHwFileSeek 36
- SceHwFileTell 37
- SceHwMemoryAlignAlloc 38
- SceHwMemoryAlloc 39
- SceHwMemoryFree 40
- SceHwSfxCallbackFunction 41
- SceHwSourceRuntimeStreamCallbackFunction 42
- SceHwSystemCallbackFunction 43

Macros **44**
- SCE_HEATWAVE_API_FAILED 45
- SCE_HEATWAVE_API_SUCCEEDED 46
- SCE_HEATWAVE_DEPRECATED 47

Document serial number: 000010343062

000010343062

©SIE Inc

Functions **48**

sceHeatWaveCloseUserOutput 49
sceHeatWaveCreateSound 50
sceHeatWaveGetInfo 51
sceHeatWaveGetSound 52
sceHeatWaveGetSoundCount 53
sceHeatWaveGetSupportedMimeTypes 54
sceHeatWaveGetVersion 55
sceHeatWaveInit 56
sceHeatWaveOpenUserOutput 57
sceHeatWaveRelease 58
sceHeatWaveReleaseSfx 59
sceHeatWaveReleaseSound 60
sceHeatWaveReleaseSource 61
sceHeatWaveRender 62
sceHeatWaveSetCallback 63
sceHeatWaveSetFileHandlers 64
sceHeatWaveSetListener 65
sceHeatWaveSetMemoryHandlers 66
sceHeatWaveSetSfxCallback 67
sceHeatWaveSfxGetInfo 68
sceHeatWaveSfxGetPitchMultiplier 69
sceHeatWaveSfxGetState 70
sceHeatWaveSfxGetVolume 71
sceHeatWaveSfxPause 72
sceHeatWaveSfxPlay 73
sceHeatWaveSfxResume 74
sceHeatWaveSfxSetForward 75
sceHeatWaveSfxSetPitchMultiplier 76
sceHeatWaveSfxSetPosition 77
sceHeatWaveSfxSetVelocity 78
sceHeatWaveSfxSetVolume 79
sceHeatWaveSfxStop 80
sceHeatWaveSoundCreateSfx 81
sceHeatWaveSoundCreateSource 82
sceHeatWaveSoundGetInfo 83
sceHeatWaveSoundGetSfx 84
sceHeatWaveSoundGetSfxCount 85
sceHeatWaveSoundGetSource 86
sceHeatWaveSoundGetSourceCount 87
sceHeatWaveSoundSetLoopCount 88
sceHeatWaveSourceGetInfo 89
sceHeatWaveSourceLoadFile 90
sceHeatWaveSourceLoadMemory 91
sceHeatWaveSourceLoadRuntimeStream 92
sceHeatWaveSourceSetCone 93
sceHeatWaveSourceSetLoopCount 95

Document serial number: 000010343062

000010343062

©SIE Inc

sceHeatWaveSourceSetPan....96
sceHeatWaveSourceSetPitchMultiplier....97
sceHeatWaveSourceSetRolloffDistances....98
sceHeatWaveSourceSetType....99
sceHeatWaveSourceSetVolume....100
Constants....101
Constant Summary....102
Defines....103
Define Summary....104

000010343062

Document serial number: 000010343062

Data Types

000010343062

Document serial number: 000010343062

©SIE Inc

float32_t

Defines a 32-bit float.

Definition

```
#include <heatwave_api_types.h>
typedef float float32_t;
```

Description

Defines a 32-bit float.

000010343062

Document serial number: 000010343062

SceHwError

Defines a HeatWave error type. This is used as the return type by all HeatWave API functions.

Definition

```
#include <heatwave_api_types.h>
typedef int32_t SceHwError;
```

Description

Defines a HeatWave error type. This is used as the return type by all HeatWave API functions.

Notes

An `SceHwError` could hold a result indicating success.

000010343062

Document serial number: 000010343062

SceHwFileFunctions

Represents user overrides for the file operation functions that the HeatWave library uses.

Definition

```
#include <heatwave_api_types.h>
typedef struct SceHwFileFunctions {
    SceHwFileOpen pOpen;
    SceHwFileClose pClose;
    SceHwFileSeek pSeek;
    SceHwFileTell pTell;
    SceHwFileRead pRead;
} SceHwFileFunctions;
```

Members

pOpen	A user defined version of `fopen()`.
pClose	A user defined version of `fclose()`.
pSeek	A user defined version of `fseek()`.
pTell	A user defined version of `ftell()`.
pRead	A user defined version of `fread()`.

Description

Represents user overrides for the file operation functions that the HeatWave library uses. Each member of a `SceHwFileFunctions` object must point to a valid custom function before the object is passed as an argument to `sceHeatWaveSetFileHandlers()`.

See Also

`sceHeatWaveSetFileHandlers()`

000010343062

Document serial number: 000010343062

SceHwHandle

Base HeatWave API handle type.

Definition

```
#include <heatwave_api_types.h>
typedef uint64_t SceHwHandle;
```

Description

Base HeatWave API handle type. All handles used by the HeatWave API derive from `SceHwHandle`.

000010343062

Document serial number: 000010343062

SceHwInfo

Represents information about HeatWave.

Definition

```
#include <heatwave_api_types.h>
typedef struct SceHwInfo {
    SceHwSystemOutputType eOutputType;
    bool bSupportedPorts[SCE_HEATWAVE_API_OUTPUT_MAX];
} SceHwInfo;
```

Members

eOutputType	The type of audio output in use.
bSupportedPorts	The currently supported audio output destinations.

Description

Represents information about HeatWave.

See Also

sceHeatWaveGetInfo()

000010343062

Document serial number: 000010343062

SceHwInitParams

Represents the initialization settings for HeatWave library.

Definition

```
#include <heatwave_api_types.h>
typedef struct SceHwInitParams {
    bool bLoopBack;
} SceHwInitParams;
```

Members

bLoopBack	A flag that specifies whether to use loopback mode. When in this mode, the Heatwave library will not feed data directly into the audio output, and sceHeatWaveRender() must be called to render the data into a suitable buffer.

Description

Represents the initialization settings for HeatWave library.

Notes

Loopback mode is not supported on the Windows platform.

See Also

sceHeatWaveInit()

0000103430 62

Document serial number: 000010343062

©SIE Inc

SceHwMemoryFileInfo

Represents information about a memory location in which an audio file has been loaded.

Definition

```
#include <heatwave_api_types.h>
typedef struct SceHwMemoryFileInfo {
    char *pBuffer;
    uint32_t uBufferLen;
} SceHwMemoryFileInfo;
```

Members

pBuffer	A pointer to the memory containing the audio data.
uBufferLen	The length in bytes of *pBuffer*.

Description

Represents information about a memory location in which an audio file has been loaded. Before a call is made to sceHeatWaveSourceLoadMemory(), memory must be allocated for the *pBuffer* member and then filled with the contents of an audio file.

Currently, uncompressed 16-bit WAV files are the only supported format.

See Also

sceHeatWaveSourceLoadMemory()

000010343062

Document serial number: 000010343062

©SIE Inc

SceHwMemoryFunctions

Represents user-provided memory functions for HeatWave to use.

Definition

```
#include <heatwave_api_types.h>
typedef struct SceHwMemoryFunctions {
    SceHwMemoryAlloc pAlloc;
    SceHwMemoryAlignAlloc pAlignedAlloc;
    SceHwMemoryFree pFree;
} SceHwMemoryFunctions;
```

Members

pAlloc	A user defined memory allocation function.
pAlignedAlloc	A user defined aligned memory allocation function.
pFree	A user defined memory freeing function.

Description

Represents user-provided memory functions for HeatWave to use. Each member of a `SceHwMemoryFunctions` object must point to a valid custom function before the object is passed as an argument to `sceHeatWaveSetMemoryHandlers()`.

See Also

`sceHeatWaveSetMemoryHandlers()`

0000103430 62

Document serial number: 000010343062

©SIE Inc

SceHwSfx

Defines a **sound effect** (an instance of a sound) to use for playback.

Definition

```
#include <heatwave_api_types.h>
typedef SceHwHandle SceHwSfx;
```

Description

Defines a **sound effect** (an instance of a sound) to use for playback. A sound effect needs to be created before any audio can actually be heard and manipulated (setting the volume of the audio, for example). Multiple sound effects can reference a single sound and these sound effects could all play that sound simultaneously.

See Also

sceHeatWaveSoundCreateSfx()

000010343062

Document serial number: 000010343062

SceHwSfxCallbackInfo

Information structure passed to user callback function.

Definition

```
#include <heatwave_api_types.h>
typedef struct SceHwSfxCallbackInfo {
    SceHwSfx hSfxHandle;
    SceHwSfxState currentState;
    SceHwSfxState previousState;
} SceHwSfxCallbackInfo;
```

Members

hSfxHandle	The `SceHwSfx` which triggered the callback.
currentState	Current state of `hSfxHandle`.
previousState	Previous state of `hSfxHandle`.

Description

Information structure passed to user callback function.

See Also

`sceHeatWaveSetSfxCallback()`

000010343062

Document serial number: 000010343062

SceHwSfxCallbackType

Defines triggers of an Sfx callback.

Definition

```
#include <heatwave_api_types.h>
typedef enum SceHwSfxCallbackType {
    eSfxCbType_StateChange = 0
} SceHwSfxCallbackType;
```

Enumeration Values

Macro	Value	Description
eSfxCbType_StateChange	0	The state of the SceHwSfx has changed.

Description

Defines triggers of an Sfx callback. Indicates the reason for execution of SceHwSfxCallbackFunction.

See Also

sceHeatWaveSetSfxCallback()

0000103430 62

Document serial number: 000010343062

©SIE Inc

SceHwSfxInfo

Represents information about a sound effect.

Definition

```
#include <heatwave_api_types.h>
typedef struct SceHwSfxInfo {
    SceHwSound hParentSound;
    SceHwVector vPosition;
    SceHwVector vVelocity;
    SceHwVector vForward;
    SceHwSfxState eSfxState;
} SceHwSfxInfo;
```

Members

hParentSound	The sound that the sound effect references.
vPosition	The current 3D position.
vVelocity	The current 3D velocity.
vForward	The current 3D forward direction.
eSfxState	The current playback state.

Description

Represents information about a sound effect.

See Also

sceHeatWaveSfxGetInfo()

000010343062

Document serial number: 000010343062

SceHwSfxPlayParam

Information for playback as an Audio3d object.

Definition

```
#include <heatwave_api_types.h>
typedef struct SceHwSfxPlayParam {
    bool bIsAudio3d;
    int32_t nAudio3dPriority;
    int32_t nDestination;
} SceHwSfxPlayParam;
```

Members

bIsAudio3d	DEPRECATED from v1.4 - *nDestination* must be used instead.
nAudio3dPriority	Priority of the SceHwSfx Audio3d object.
nDestination	The output destination for the SceHwSfx. Must be one of the SCE_HEATWAVE_API_OUTPUT_XXX values.

Description

Information for playback as an Audio3d object.

See Also

sceHeatWaveSfxPlay()

000010343062

Document serial number: 000010343062

SceHwSfxState

Defines the possible playback states of a sound effect.

Definition

```
#include <heatwave_api_types.h>
typedef enum SceHwSfxState {
    kSfxState_Ready = 0,
    kSfxState_Playing = 1,
    kSfxState_Paused = 2
} SceHwSfxState;
```

Enumeration Values

Macro	Value	Description
kSfxState_Ready	0	The sound effect is ready to be played.
kSfxState_Playing	1	The sound effect is outputting audio data.
kSfxState_Paused	2	The sound effect has been paused.

Description

Defines the possible playback states of a sound effect.

See Also

sceHeatWaveSfxGetState()

000010343062

Document serial number: 000010343062

SceHwSound

Defines a single **sound**, which is a collection of sound sources that are played together.

Definition

```
#include <heatwave_api_types.h>
typedef SceHwHandle SceHwSound;
```

Description

Defines a single **sound**, which is a collection of sound sources that are played together. A sound can be referenced by multiple sound effects, which enables simultaneously occurring events to all make use of a single sound. In such cases, an instance of a sound being played can overlap in time with other instances of it being played.

Because a sound can play several sound sources in unison, it also enables more than one sound source to be heard in response to a particular event occurring. The sound sources belonging to a sound can be a mixture of 2D and 3D.

See Also

sceHeatWaveCreateSound()

000010343062

Document serial number: 000010343062

SceHwSoundInfo

Represents information about a sound.

Definition

```
#include <heatwave_api_types.h>
typedef struct SceHwSoundInfo {
    uint32_t uNumSfx;
    uint32_t uNumSources;
} SceHwSoundInfo;
```

Members

uNumSfx	The number of sound effects that reference the sound.
uNumSources	The number of sound sources that belong to the sound.

Description

Represents information about a sound.

See Also

sceHeatWaveSoundGetInfo()

000010343062

Document serial number: 000010343062

©SIE Inc

SceHwSource

Defines a **sound source**.

Definition

```
#include <heatwave_api_types.h>
typedef SceHwHandle SceHwSource;
```

Description

Defines a **sound source**. This will be an audio file. Emitter cones and volume roll-off distances can be set per sound source. Currently, uncompressed 16 bit WAV files are the only supported audio file format.

See Also

sceHeatWaveSoundCreateSource()

000010343062

Document serial number: 000010343062

SceHwSourceInfo

Represents information about a sound source.

Definition

```
#include <heatwave_api_types.h>
typedef struct SceHwSourceInfo {
    SceHwSound hParentSound;
    int32_t nLoopCount;
    SceHwSourceType eType;
    float32_t fGain;
    float32_t fPitch;
} SceHwSourceInfo;
```

Members

hParentSound	The parent sound that the sound source belongs to.
nLoopCount	The number of times the audio data is set to loop on playback.
eType	Specifies whether this sound source is a 2D or 3D sound emitter.
fGain	The volume multiplier of the source.
fPitch	The pitch multiplier of the source.

Description

Represents information about a sound source.

See Also

sceHeatWaveSourceGetInfo()

000010343062

Document serial number: 000010343062

©SIE Inc

SceHwSourceRuntimeStreamCallbackInfo

Information structure passed to `SceHwSourceRuntimeStreamCallbackFunction`.

Definition

```
#include <heatwave_api_types.h>
typedef struct SceHwSourceRuntimeStreamCallbackInfo {
    char *pcBuffer;
    int32_t nBufferSizeBytes;
    int32_t nBytesWrittenOut;
} SceHwSourceRuntimeStreamCallbackInfo;
```

Members

pcBuffer	Buffer to receive audio data.
nBufferSizeBytes	Size of buffer in bytes.
nBytesWrittenOut	The amount of data written to the buffer by the callback.

Description

Information structure passed to `SceHwSourceRuntimeStreamCallbackFunction`.

See Also

`sceHeatWaveSourceLoadRuntimeStream()`

000010343062

Document serial number: 000010343062

SceHwSourceRuntimeStreamParam

Information about a runtime stream.

Definition

```
#include <heatwave_api_types.h>
typedef struct SceHwSourceRuntimeStreamParam {
    int32_t nChannelCount;
    int32_t nSampleRate;
} SceHwSourceRuntimeStreamParam;
```

Members

nChannelCount	The channel count of the audio data.
nSampleRate	The sample rate of the audio data.

Description

Information about a runtime stream.

See Also

sceHeatWaveSourceLoadRuntimeStream()

0000103430 62

Document serial number: 000010343062

SceHwSourceType

Defines the ways in which a sound source can be played.

Definition

```
#include <heatwave_api_types.h>
typedef enum SceHwSourceType {
    eSource2d = 0,
    eSource3d = 1
} SceHwSourceType;
```

Enumeration Values

Macro	Value	Description
eSource2d	0	Play the sound source as a 2D sound at the listener position.
eSource3d	1	Play the sound source with 3D positional information.

Description

Defines the ways in which a sound source can be played. It is possible to pan 2D sound sources while 3D sound sources can have a sound cone and roll-off distance set. The sound sources belonging to a sound can be a mixture of 2D and 3D.

See Also

sceHeatWaveSourceGetInfo(), sceHeatWaveSourceSetType(), sceHeatWaveSourceSetPan(), sceHeatWaveSourceSetCone(), sceHeatWaveSourceSetRolloffDistances()

000010343062

Document serial number: 000010343062

©SIE Inc

SceHwSystemCallbackInfo

Information structure passed to user callback function.

Definition

```
#include <heatwave_api_types.h>
typedef struct SceHwSystemCallbackInfo {
    uint32_t uCurrentA3dObjectCount;
    uint32_t uPreviousA3dObjectCount;
} SceHwSystemCallbackInfo;
```

Members

uCurrentA3dObjectCount	Number of currently active libAudio3d objects.
uPreviousA3dObjectCount	Previous number of active libAudio3d objects.

Description

Information structure passed to user callback function.

See Also

sceHeatWaveSetCallback()

000010343062

Document serial number: 000010343062

SceHwSystemCallbackType

Defines triggers of the system callback.

Definition

```
#include <heatwave_api_types.h>
typedef enum SceHwSystemCallbackType {
    eSysCbType_A3dObjectCount = 0
} SceHwSystemCallbackType;
```

Enumeration Values

Macro	Value	Description
eSysCbType_A3dObjectCount	0	The number of active Audio3d objects has changed.

Description

Defines triggers of the system callback. Indicates the reason for execution of SceHwSystemCallbackFunction.

See Also

sceHeatWaveSetCallback()

0000103430 62

Document serial number: 000010343062

©SIE Inc

SceHwSystemOutputType

Defines the audio outputs that HeatWave can use.

Definition

```
#include <heatwave_api_types.h>
typedef enum SceHwSystemOutputType {
    eSysOutput_Ngs = 0,
    eSysOutput_Multistream = 1,
    eSysOutput_Ngs2 = 2,
    eSysOutput_XAudio2 = 3,
    eSysOutput_Audio3d = 4,
    eSysOutput_UserRender = 5
} SceHwSystemOutputType;
```

Enumeration Values

Macro	Value	Description
eSysOutput_Ngs	0	PlayStation®Vita NGS Synth.
eSysOutput_Multistream	1	PlayStation®3 Multistream.
eSysOutput_Ngs2	2	PlayStation®4 NGS2 Synth.
eSysOutput_XAudio2	3	Win XAudio.
eSysOutput_Audio3d	4	PlayStation®4 Audio3d.
eSysOutput_UserRender	5	Output is performed via sceHeatWaveRender().

Description

Defines the audio outputs that HeatWave can use.

See Also

sceHeatWaveGetInfo()

0000103430 62

Document serial number: 000010343062

©SIE Inc

SceHwVector

Defines a coordinate or vector in 3D space.

Definition

```
#include <heatwave_api_types.h>
typedef struct SceHwVector {
    float32_t fX;
    float32_t fY;
    float32_t fZ;
} SceHwVector;
```

Members

fX	The x axis component.
fY	The y axis component.
fZ	The z axis component.

Description

Defines a coordinate or vector in 3D space.

See Also

sceHeatWaveSetListener(), sceHeatWaveSfxSetPosition(), sceHeatWaveSfxSetVelocity(), sceHeatWaveSfxSetForward()

000010343062

Document serial number: 000010343062

ScewHwSourceRuntimeStreamCallbackType

Defines the triggers of the `SceHwSource` runtime audio callback.

Definition

```
#include <heatwave_api_types.h>
typedef enum ScewHwSourceRuntimeStreamCallbackType {
    eStreamCreate = 0,
    eStreamRelease,
    eStreamRender,
    eStreamReset
} ScewHwSourceRuntimeStreamCallbackType;
```

Enumeration Values

Macro	Value	Description
`eStreamCreate`	`0`	Create an instance of the stream.
`eStreamRelease`	N/A	Release an instance of the stream.
`eStreamRender`	N/A	Render a buffer of audio data.
`eStreamReset`	N/A	Reset the stream to initial state.

Description

Defines the triggers of the `SceHwSource` runtime audio callback. Defines the reason for the execution of `SceHwSystemCallbackFunction`.

See Also

`sceHeatWaveSourceLoadRuntimeStream`

000010343062

Document serial number: 000010343062

Callback Functions

000010343062

Document serial number: 000010343062

©SIE Inc

SceHwFileClose

Defines a custom `fclose()` handler.

Definition

```
#include <heatwave_api_types.h>
typedef int32_t (*SceHwFileClose)(
    void *pStream
);
```

Arguments

pStream	The file stream to close.

Return Values

A value of 0 is returned if the operation was successful.

Description

Defines a custom `fclose()` handler.

See Also

SceHwFileOpen()

0000103430 62

Document serial number: 000010343062

SceHwFileOpen

Defines a custom `fopen()` handler.

Definition

```
#include <heatwave_api_types.h>
typedef void * (*SceHwFileOpen)(
    const char *strFilePath,
    const char *strMode
);
```

Arguments

strFilePath	The full path of the file to open.
strMode	The mode to open the file in.

Return Values

A pointer to the opened file if the operation was successful; otherwise `NULL` is returned.

Description

Defines a custom `fopen()` handler.

See Also

`SceHwFileClose()`

000010343062

Document serial number: 000010343062

SceHwFileRead

Defines a custom `fread()` handler.

Definition

```
#include <heatwave_api_types.h>
typedef uint32_t (*SceHwFileRead)(
    void *pPtr,
    uint32_t uSize,
    uint32_t uCount,
    void *pStream
);
```

Arguments

pPtr	The memory block to read data into.
uSize	The size of an element in bytes.
uCount	The number of elements to read.
pStream	The file stream to read the elements from.

Return Values

The number of elements read.

Description

Defines a custom `fread()` handler.

See Also

`SceHwFileOpen()`, `SceHwFileSeek()`

0000103430 62

Document serial number: 000010343062

SceHwFileSeek

Defines a custom `fseek()` handler.

Definition

```
#include <heatwave_api_types.h>
typedef int32_t (*SceHwFileSeek)(
    void *pStream,
    uint32_t uOffset,
    int32_t iOrigin
);
```

Arguments

pStream	The file stream to seek within.
uOffset	The number of bytes to offset from *iOrigin*.
iOrigin	The position in the file from which offset is applied. This can be set to *SEEK_SET*, *SEEK_CUR* or *SEEK_END*.

Return Values

0 A value of 0 is returned if the operation was successful.

Description

Defines a custom `fseek()` handler.

See Also

`SceHwFileOpen()`, `SceHwFileTell()`

0000103430 62

Document serial number: 000010343062

SceHwFileTell

Defines a custom file `ftell()` handler.

Definition

```
#include <heatwave_api_types.h>
typedef int32_t (*SceHwFileTell)(
    void *pStream
);
```

Arguments

pStream	The file stream to report the position of.

Return Values

The current position of *pStream*. A value of -1L indicates an error has occurred.

Description

Defines a custom file `ftell()` handler.

See Also

SceHwFileOpen(), SceHwFileSeek()

0000103430 62

Document serial number: 000010343062

SceHwMemoryAlignAlloc

Defines an aligned memory allocation function.

Definition

```
#include <heatwave_api_types.h>
typedef void * (*SceHwMemoryAlignAlloc)(
    size_t align,
    size_t size
);
```

Arguments

align	The alignment of the memory to allocate.
size	The size of the memory to allocate.

Return Values

A pointer to the newly allocated memory if the operation was successful; otherwise `NULL` is returned.

Description

Defines an aligned memory allocation function.

See Also

SceHwMemoryFree()

000010343062

Document serial number: 000010343062

SceHwMemoryAlloc

Defines a memory allocation function.

Definition

```
#include <heatwave_api_types.h>
typedef void * (*SceHwMemoryAlloc)(
    size_t size
);
```

Arguments

size	The size of the memory to allocate.

Return Values

A pointer to the newly allocated memory if the operation was successful; otherwise `NULL` is returned.

Description

Defines a memory allocation function.

See Also

`SceHwMemoryFree()`

0000103430 62

Document serial number: 000010343062

©SIE Inc

SceHwMemoryFree

Defines a memory freeing function.

Definition

```
#include <heatwave_api_types.h>
typedef void (*SceHwMemoryFree)(
    void *ptr
);
```

Arguments

ptr	A pointer to a previously allocated memory block.

Return Values

None

Description

Defines a memory freeing function.

See Also

SceHwMemoryAlloc()

000010343062

Document serial number: 000010343062

SceHwSfxCallbackFunction

Defines the user callback function.

Definition

```
#include <heatwave_api_types.h>
typedef void (*SceHwSfxCallbackFunction)(
    const SceHwSfxCallbackInfo *pCallbackInfo,
    void *pUserData,
    SceHwSfxCallbackType reason
);
```

Arguments

pCallbackInfo	Callback information.
pUserData	The user data set by sceHeatWaveSetSfxCallback().
reason	SceHwSfxCallbackType specifying why the callback is being issued.

Return Values

None

Description

Defines the user callback function.

See Also

sceHeatWaveSetSfxCallback()

000010343062

Document serial number: 000010343062

SceHwSourceRuntimeStreamCallbackFunction

Defines the user callback used to generate audio data at runtime.

Definition

```
#include <heatwave_api_types.h>
typedef void (*SceHwSourceRuntimeStreamCallbackFunction)(
    SceHwSourceRuntimeStreamCallbackInfo *pCallbackInfo,
    void *pUserData,
    ScewHwSourceRuntimeStreamCallbackType reason
);
```

Arguments

pCallbackInfo	Callback information. The `nSampleRate` and `nChannelCount` should be set by the user when *reason* is `eStreamCreate`.
pUserData	The user data set by `sceHeatWaveSourceLoadRuntimeStream()`.
reason	`ScewHwSourceRuntimeStreamCallbackType` specifying why the callback was issued.

Return Values

None

Description

Defines the user callback used to generate audio data at runtime.

See Also

`sceHeatWaveSourceLoadRuntimeStream()`

000010343062

Document serial number: 000010343062

©SIE Inc

SceHwSystemCallbackFunction

Defines the user callback function.

Definition

```
#include <heatwave_api_types.h>
typedef void (*SceHwSystemCallbackFunction)(
    const SceHwSystemCallbackInfo *pCallbackInfo,
    void *pUserData,
    SceHwSystemCallbackType reason
);
```

Arguments

pCallbackInfo	Callback information.
pUserData	The user data set by sceHeatWaveSetCallback().
reason	SceHwSystemCallbackType specifying why the callback is being issued.

Return Values

None

Description

Defines the user callback function.

See Also

sceHeatWaveSetCallback()

000010343062

Document serial number: 000010343062

Macros

000010343062

Document serial number: 000010343062

©SIE Inc

SCE_HEATWAVE_API_FAILED

Checks if an operation was **not** successful.

Definition

```
#include <heatwave_api_errors.h>
#define SCE_HEATWAVE_API_FAILED(
    res
) (res != SCE_HEATWAVE_API_OK)
```

Arguments

res	The return code to check, which should be an `SceHwError`.

Return Values

A value of true if `res` is an error code; otherwise false is returned.

Description

Checks if an operation was **not** successful.

0000103430 62

Document serial number: 000010343062

SCE_HEATWAVE_API_SUCCEEDED

Checks if an operation was successful.

Definition

```
#include <heatwave_api_errors.h>
#define SCE_HEATWAVE_API_SUCCEEDED(
    res
) (res == SCE_HEATWAVE_API_OK)
```

Arguments

res	The return code to check, which should be an `SceHwError`.

Return Values

A value of true if `res` is `SCE_HEATWAVE_API_OK`; otherwise false is returned.

Description

Checks if an operation was successful.

000010343062

Document serial number: 000010343062

SCE_HEATWAVE_DEPRECATED

Used to mark HeatWave API functions as deprecated.

Definition

```
#include <heatwave_api.h>
#define SCE_HEATWAVE_DEPRECATED(func) func __attribute__ ((deprecated))
```

Arguments

None

Return Values

None

Description

Used to mark HeatWave API functions as deprecated. Use of deprecated functions should be avoided as they will be removed in later versions of the library.

000010343062

Document serial number: 000010343062

Functions

000010343062

Document serial number: 000010343062

©SIE Inc

sceHeatWaveCloseUserOutput

Close a user-specific audio output.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveCloseUserOutput(
    int32_t nOutputId
);
```

Arguments

nOutputId	SCE_HEATWAVE_API_OUTPUT_PAD_XXX.

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Close a user-specific audio output. Close a previously opened user-specific audio output.

000010343062

Document serial number: 000010343062

sceHeatWaveCreateSound

Creates a new sound.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveCreateSound(
    SceHwSound *pOutSoundHandle
);
```

Arguments

[out] *pOutSoundHandle*	Receives a handle to the newly created sound if the operation was successful.

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Creates a new sound. At least one sound must be created before any sound effects can be created. When a sound is no longer needed, the resources it uses should be released by calling sceHeatWaveReleaseSound().

See Also

sceHeatWaveReleaseSound(), sceHeatWaveGetSound(), sceHeatWaveGetSoundCount()

000010343062

Document serial number: 000010343062

sceHeatWaveGetInfo

Gets global information about HeatWave.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveGetInfo(
    SceHwInfo *pInfoOut
);
```

Arguments

[out] *pInfoOut* Receives the information about HeatWave.

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Gets global information about HeatWave.

See Also

SceHwInfo

0000103430 62

Document serial number: 000010343062

sceHeatWaveGetSound

Gets an existing sound by index.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveGetSound(
    int32_t nIndex,
    SceHwSound *pOutSoundHandle
);
```

Arguments

[in] *nIndex*	The index of the sound to retrieve.
[out] *pOutSoundHandle*	Receives the sound.

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Gets an existing sound by index. The sounds currently existing in HeatWave are held in a zero-based array.

See Also

sceHeatWaveGetSoundCount(), sceHeatWaveCreateSound(), sceHeatWaveReleaseSound()

000010343062

Document serial number: 000010343062

sceHeatWaveGetSoundCount

Gets the number of sounds that exist.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveGetSoundCount(
    int32_t *pOutCount
);
```

Arguments

[out] *pOutCount*	Receives the number of sounds.

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Gets the number of sounds that exist. This function can be used in conjunction with sceHeatWaveGetSound() to loop through all the sounds currently existing in HeatWave.

See Also

sceHeatWaveGetSound(), sceHeatWaveCreateSound(), sceHeatWaveReleaseSound()

0000103430 62

Document serial number: 000010343062

sceHeatWaveGetSupportedMimeTypes

Gets a list of supported audio formats as MIME type strings.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveGetSupportedMimeTypes(
    const char **ppMimeListOut[],
    uint32_t *pSizeOut
);
```

Arguments

[out] *ppMimeListOut*	Array of MIME types as strings
[out] *pSizeOut*	Number of entries in the array

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Gets a list of supported audio formats as MIME type strings.

000010343062

Document serial number: 000010343062

sceHeatWaveGetVersion

Gets the version identifier of the HeatWave library.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveGetVersion(
    uint32_t *uVersion
);
```

Arguments

[out] *uVersion* Receives the version identifier.

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Gets the version identifier of the HeatWave library.

See Also

SCE_HEATWAVE_API_VERSION

0000103430 62

Document serial number: 000010343062

©SIE Inc

sceHeatWaveInit

Initializes the HeatWave library.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveInit(
    SceHwInitParams *pParams
);
```

Arguments

[in] *pParams* The initialization parameters.

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Initializes the HeatWave library.

Notes

This function **must** be called before any other HeatWave function except for sceHeatWaveSetMemoryHandlers() and sceHeatWaveSetFileHandlers().

See Also

sceHeatWaveRelease(), sceHeatWaveSetMemoryHandlers(), sceHeatWaveSetFileHandlers()

000010343062

Document serial number: 000010343062

sceHeatWaveOpenUserOutput

Open a user-specific user audio output.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveOpenUserOutput(
    int32_t nOutputId,
    int32_t nUserServiceID
);
```

Arguments

nOutputId	SCE_HEATWAVE_API_OUTPUT_PAD_XXX.
nUserServiceID	The platform handle representing the user service ID.

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Open a user-specific user audio output. Open the audio output destination for the specified user handle.

000010343062

Document serial number: 000010343062

sceHeatWaveRelease

Releases all resources used by the HeatWave library.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveRelease(void);
```

Arguments

None

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Releases all resources used by the HeatWave library. This function releases all sounds, sound sources and sound effects that have not been released at the point it was called.

See Also

sceHeatWaveInit()

000010343062

Document serial number: 000010343062

sceHeatWaveReleaseSfx

Releases an existing sound effect.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveReleaseSfx(
    SceHwSfx hSfxHandle
);
```

Arguments

[in] *hSfxHandle*	The sound effect to be released.

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Releases an existing sound effect. Calling this function frees up the resources used by the sound effect and stops it playing immediately.

See Also

sceHeatWaveSoundCreateSfx(), sceHeatWaveSoundGetSfx(), sceHeatWaveSoundGetSfxCount()

0000103430 62

Document serial number: 000010343062

©SIE Inc

sceHeatWaveReleaseSound

Releases an existing sound.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveReleaseSound(
    SceHwSound hSoundHandle
);
```

Arguments

[in] *hSoundHandle* The sound to release.

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Releases an existing sound. Calling this function frees up the resources used by the sound. This includes all sound effects referencing a sound and all sound sources belonging to a sound. Internally, this will be done with calls to sceHeatWaveReleaseSfx() and sceHeatWaveReleaseSource().

See Also

sceHeatWaveCreateSound(), sceHeatWaveGetSound(), sceHeatWaveGetSoundCount()

000010343062

Document serial number: 000010343062

sceHeatWaveReleaseSource

Releases an existing sound source.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveReleaseSource(
    SceHwSource hSourceHandle
);
```

Arguments

[in] *hSourceHandle* The audio source to be released.

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Releases an existing sound source. Calling this function frees up the resources used by the sound source. A sound source that has been released will not be played anymore when the sound it belonged to is played by a sound effect.

See Also

sceHeatWaveSoundCreateSource(), sceHeatWaveSoundGetSource(), sceHeatWaveSoundGetSourceCount()

000010343062

Document serial number: 000010343062

sceHeatWaveRender

Render a block of audio data into a buffer.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveRender(
    int16_t *pBuffer,
    uint32_t uNumSamplesPerChannel,
    uint32_t uNumChannels
);
```

Arguments

[in] *pBuffer*	The memory area to render the audio data to.
[in] *uNumSamplesPerChannel*	The number of audio samples to render for each channel.
[in] *uNumChannels*	The number of channels.

Return Values

See SCE_HEATWAVE_API_OK and other return codes. A value of SCE_HEATWAVE_API_ERROR_OPERATION is returned if HeatWave is not in loopback mode.

Description

Render a block of audio data into a buffer. This function will only work when the Heatwave library is in loopback mode and needs to be called at a rate suitable for the destination the audio data is being rendered to. It can be used, for example, when there is a requirement to write the audio data to a file.

See Also

sceHeatWaveInit()

000010343062

Document serial number: 000010343062

sceHeatWaveSetCallback

Set the user callback for HeatWave.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveSetCallback(
    SceHwSystemCallbackFunction pCallbackFunc,
    void *pUserData = NULL
);
```

Arguments

[in] *pCallbackFunc*	The user function to execute.
[in] *pUserData*	User specific data passed to the callback function (optional).

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Set the user callback for HeatWave. The callback reports changes in the internal state of HeatWave. SceHwSystemCallbackType specifies the conditions that will cause this callback to be executed.

Notes

A value of SCE_HEATWAVE_API_NO_CALLBACK for pCallbackFunc indicates the callback should be disabled.

pUserData is optional, a value of NULL indicates no user data should be passed when executing pCallbackFunc.

See Also

SceHwSystemCallbackType, SceHwSystemCallbackFunction

Document serial number: 000010343062

000010343062

sceHeatWaveSetFileHandlers

Overrides the default file operation functions that HeatWave uses.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveSetFileHandlers(
    SceHwFileFunctions *pUserOverrides
);
```

Arguments

[in] *pUserOverrides*	The function pointers to the user defined functions, which will override the defaults.

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Overrides the default file operation functions that HeatWave uses.

Notes

This function **must** be called before sceHeatWaveInit().

000010343062

Document serial number: 000010343062

sceHeatWaveSetListener

Sets the listener in 3D space.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveSetListener(
    SceHwVector vPosition,
    SceHwVector vVelocity,
    SceHwVector vForward,
    SceHwVector vUp
);
```

Arguments

[in] *vPosition*	The 3D world position of the listener.
[in] *vVelocity*	The velocity of the listener, which is used for applying Doppler shift. Because this argument is not used to move the listener, it can be set to a greater or lesser value than the listener is actually moving at. This can be used to create a more or less extreme Doppler shift as required. Setting this argument to 0 means no Doppler shift will be applied unless a sound effect has a velocity of greater than 0.
[in] *vForward*	The direction the listener is facing.
[in] *vUp*	The normal to the direction vector, which is used to determine what is left and what is right for the listener.

Return Values

See `SCE_HEATWAVE_API_OK` and other return codes.

Description

Sets the listener in 3D space. Typically the listener is set to the same position, velocity and orientation as the game camera and can be thought of as a "virtual microphone". The HeatWave library only supports one listener, which is always available once the library has been initialized. This function is used to update the listener's position, velocity and rotation orientation in the game as required.

Notes

A right hand coordinate system is used.

The value passed to the *vVelocity* parameter **must** be less than `kfSpeedOfSoundSq`.

See Also

SceHwVector

Document serial number: 000010343062

000010343062

sceHeatWaveSetMemoryHandlers

Overrides the default memory allocation and deallocation functions that HeatWave uses.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveSetMemoryHandlers(
    SceHwMemoryFunctions *pUserOverrides
);
```

Arguments

[in] *pUserOverrides*	The function pointers to the user defined functions, which will override the defaults.

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Overrides the default memory allocation and deallocation functions that HeatWave uses.

Notes

This function **must** be called before sceHeatWaveInit().

000010343062

Document serial number: 000010343062

sceHeatWaveSetSfxCallback

Set the user callback for SceHwSfx.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveSetSfxCallback(
    SceHwSfxCallbackFunction pCallbackFunc,
    void *pUserData = NULL
);
```

Arguments

[in] *pCallbackFunc*	The user function to execute
[in] *pUserData*	User specific data passed to the callback function (optional)

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Set the user callback for SceHwSfx. The callback reports changes in the internal state of SceHwSfx. SceHwSfxCallbackInfo specifies the conditions that will cause the callback to be executed. The callback function is common to all SceHwSfx.

Notes

A value of SCE_HEATWAVE_API_NO_CALLBACK for pCallbackFunc indicates the callback should be disabled.

pUserData is optional, with a value of NULL indicating that no user data should be passed when executing pCallbackFunc.

See Also

SceHwSfxCallbackType, SceHwSfxCallbackInfo

Document serial number: 000010343062

000010343062

sceHeatWaveSfxGetInfo

Gets the current information about a particular sound effect.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveSfxGetInfo(
    SceHwSfx hSfxHandle,
    SceHwSfxInfo *pOutInfo
);
```

Arguments

[in] *hSfxHandle*	The sound effect to query.
[out] *pOutInfo*	Receives the current information about the sound effect if the operation was successful.

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Gets the current information about a particular sound effect. This consists of the sound that the sound effect references, the 3D position, velocity and direction, and the current playback state.

See Also

sceHeatWaveSoundCreateSfx(), sceHeatWaveSoundGetSfx(), sceHeatWaveSfxSetPosition(), sceHeatWaveSfxSetVelocity(), sceHeatWaveSfxSetForward()

000010343062

Document serial number: 000010343062

sceHeatWaveSfxGetPitchMultiplier

Gets the playback pitch of a sound effect.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveSfxGetPitchMultiplier(
    SceHwSfx hSfxHandle,
    float32_t *pOutPitchMultiplier
);
```

Arguments

[in] *hSfxHandle*	The sound effect to get the pitch multiplier for.
[out] *pOutPitchMultiplier*	Receives the pitch multiplier for the sound effect if the operation was successful.

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Gets the playback pitch of a sound effect.

See Also

sceHeatWaveSfxSetPitchMultiplier()

000010343062

Document serial number: 000010343062

sceHeatWaveSfxGetState

Gets the current state of a particular sound effect.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveSfxGetState(
    SceHwSfx hSfxHandle,
    SceHwSfxState *pOutState
);
```

Arguments

[in] *hSfxHandle*	The sound effect to query.
[out] *pOutState*	Receives the current state of the sound effect if the operation was successful.

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Gets the current state of a particular sound effect. A sound effect can either be ready to be played, playing or paused.

See Also

sceHeatWaveSfxPlay(), sceHeatWaveSfxStop(), sceHeatWaveSfxPause(), sceHeatWaveSfxResume()

000010343062

Document serial number: 000010343062

sceHeatWaveSfxGetVolume

Gets the playback volume (gain) of a sound effect.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveSfxGetVolume(
    SceHwSfx hSfxHandle,
    float32_t *pOutVolume
);
```

Arguments

[in] *hSfxHandle*	The sound effect to get the volume for.
[out] *pOutVolume*	Receives the volume at which the sound effect is playing if the operation was successful.

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Gets the playback volume (gain) of a sound effect.

See Also

sceHeatWaveSfxSetVolume()

000010343062

Document serial number: 000010343062

sceHeatWaveSfxPause

Pauses a sound effect.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveSfxPause(
    SceHwSfx hSfxHandle
);
```

Arguments

[in] *hSfxHandle* The sound effect to pause.

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Pauses a sound effect. When this function is called, all sound sources being played for this particular sound effect will stop playing, but the point at which they stopped and, where applicable, the number of loops they have completed will be remembered. If sceHeatWaveSfxResume() is called, play will continue from this point.

See Also

sceHeatWaveSfxPlay(), sceHeatWaveSfxStop(), sceHeatWaveSfxResume(), sceHeatWaveSfxGetState()

000010343062

Document serial number: 000010343062

sceHeatWaveSfxPlay

Plays a sound effect.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveSfxPlay(
    SceHwSfx hSfxHandle,
    SceHwSfxPlayParam *pParam = NULL
);
```

Arguments

[in] *hSfxHandle*	The sound effect to play.
[in] *pParam*	Settings to control playback.

Return Values

See SCE_HEATWAVE_API_OK and other return codes

Description

Plays a sound effect. This function will play all the sound sources that belong to the sound that this sound effect references.

See Also

sceHeatWaveSfxStop(), sceHeatWaveSfxPause(), sceHeatWaveSfxResume(), sceHeatWaveSfxGetState()

000010343062

Document serial number: 000010343062

sceHeatWaveSfxResume

Resumes playback of a paused sound effect.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveSfxResume(
    SceHwSfx hSfxHandle
);
```

Arguments

[in] *hSfxHandle*	The sound effect to resume playing.

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Resumes playback of a paused sound effect.

See Also

sceHeatWaveSfxPlay(), sceHeatWaveSfxStop(), sceHeatWaveSfxPause(), sceHeatWaveSfxGetState()

000010343062

Document serial number: 000010343062

sceHeatWaveSfxSetForward

Sets the direction of a sound effect in 3D space.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveSfxSetForward(
    SceHwSfx hSfxHandle,
    SceHwVector vForward
);
```

Arguments

[in] *hSfxHandle*	The sound effect to update.
[in] *vForward*	The new direction for the sound effect.

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Sets the direction of a sound effect in 3D space. This function is used to set the sound cone direction for the sound sources that will be played as part of this sound effect. Sound cones for individual sound sources are set by calling sceHeatWaveSourceSetCone().

Notes

A right hand coordinate system is used for this operation.

This function is only relevant when a sound source being played for this sound effect has been defined as 3D.

See Also

sceHeatWaveSfxSetPosition(), sceHeatWaveSfxSetVelocity(), SceHwVector, sceHeatWaveSourceSetCone()

000010343062

Document serial number: 000010343062

sceHeatWaveSfxSetPitchMultiplier

Sets the playback pitch for a sound effect.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveSfxSetPitchMultiplier(
    SceHwSfx hSfxHandle,
    float32_t fPitchMultiplier
);
```

Arguments

[in] *hSfxHandle*	The sound effect to set the pitch multiplier for.
[in] *fPitchMultiplier*	The new pitch multiplier for the sound effect. This must be a value between `kfSfxPitchMin` and `kfSfxPitchMax`.

Return Values

See `SCE_HEATWAVE_API_OK` and other return codes.

Description

Sets the playback pitch for a sound effect.

See Also

`sceHeatWaveSfxGetPitchMultiplier()`

000010343062

Document serial number: 000010343062

sceHeatWaveSfxSetPosition

Sets the 3D world position for a sound effect.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveSfxSetPosition(
    SceHwSfx hSfxHandle,
    SceHwVector vPosition
);
```

Arguments

[in] *hSfxHandle*	The sound effect to set the 3D world position to for.
[in] *vPosition*	The new 3D world coordinates for the sound effect.

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Sets the 3D world position for a sound effect. This function is used to set the x, y and z coordinates for a sound effect in 3D space. The sound cones for the sound sources that will be played as part of this sound effect will emanate from this point. Sound cones for individual sound sources are set by calling sceHeatWaveSourceSetCone().

Notes

A right hand coordinate system is used for this operation.

This function is only relevant when a sound source being played for this sound effect has been defined as 3D.

See Also

sceHeatWaveSfxSetVelocity(), sceHeatWaveSfxSetForward(), SceHwVector, sceHeatWaveSourceSetCone()

0000103430 62

Document serial number: 000010343062

©SIE Inc

sceHeatWaveSfxSetVelocity

Sets the velocity of a sound effect.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveSfxSetVelocity(
    SceHwSfx hSfxHandle,
    SceHwVector vVelocity
);
```

Arguments

[in] *hSfxHandle*	The sound effect to update.
[in] *vVelocity*	The new velocity for the sound effect.

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Sets the velocity of a sound effect. This function sets the velocity of the sound effect, which is used for applying Doppler shift. Because this value is not used to move the sound effect, it can be set to a greater or lesser value than the sound effect is actually moving at. This can be used to create a more or less extreme Doppler shift as required. Setting the velocity of the sound effect to 0 means no Doppler shift will be applied for it unless the listener has a velocity greater than 0.

Notes

A right hand coordinate system is used for this operation.

The value passed to the *vVelocity* parameter **must** be less than kfSpeedOfSoundSq.

This function is only relevant when a sound source being played for this sound effect has been defined as 3D.

See Also

sceHeatWaveSfxSetPosition(), sceHeatWaveSfxSetForward(), SceHwVector

000010343062

Document serial number: 000010343062

sceHeatWaveSfxSetVolume

Sets the playback volume (gain) of a sound effect.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveSfxSetVolume(
    SceHwSfx hSfxHandle,
    float32_t fVolume
);
```

Arguments

[in] *hSfxHandle*	The sound effect to set the volume for.
[in] *fVolume*	The new volume for the sound effect. This must be a value between `kfSfxVolumeMin` and `kfSfxVolumeMax`.

Return Values

See `SCE_HEATWAVE_API_OK` and other return codes.

Description

Sets the playback volume (gain) of a sound effect.

See Also

`sceHeatWaveSfxGetVolume()`

000010343062

Document serial number: 000010343062

sceHeatWaveSfxStop

Stops a sound effect from playing.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveSfxStop(
    SceHwSfx hSfxHandle
);
```

Arguments

[in] *hSfxHandle* The sound effect to stop.

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Stops a sound effect from playing. When this function is called, all sound sources being played for this particular sound effect will stop.

See Also

sceHeatWaveSfxPlay(), sceHeatWaveSfxPause(), sceHeatWaveSfxResume(), sceHeatWaveSfxGetState()

0000103430 62

Document serial number: 000010343062

sceHeatWaveSoundCreateSfx

Creates a sound effect that references the specified sound.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveSoundCreateSfx(
    SceHwSound hSoundHandle,
    SceHwSfx *pOutSfxHandle
);
```

Arguments

[in] *hSoundHandle*	The sound to get a sound effects handle for.
[out] *pOutSfxHandle*	Receives the newly created sound effect handle if the operation was successful.

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Creates a sound effect that references the specified sound.

See Also

sceHeatWaveReleaseSfx(), sceHeatWaveSoundGetSfx(), sceHeatWaveSoundGetSfxCount()

0000103430 62

Document serial number: 000010343062

sceHeatWaveSoundCreateSource

Creates a new sound source for a sound.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveSoundCreateSource(
    SceHwSound hSoundHandle,
    SceHwSource *pOutSourceHandle
);
```

Arguments

[in] *hSoundHandle*	The sound to add a new audio source to.
[out] *pOutSourceHandle*	Receives a handle to the newly created sound source if the operation was successful.

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Creates a new sound source for a sound. The sound source created by this function needs to be populated with data from an audio file before it will be possible to hear anything. Either sceHeatWaveSourceLoadFile() or sceHeatWaveSourceLoadMemory() should be used to do this. The sound source should be specified as 2D or 3D by calling sceHeatWaveSourceSetType(). It is also possible to set the 3D cone and roll-off distance (only for 3D sounds), the pan position (only for 2D sounds), and the loop count for the sound source.

See Also

sceHeatWaveReleaseSource(), sceHeatWaveSoundGetSource(), sceHeatWaveSoundGetSourceCount(), sceHeatWaveSourceLoadFile(), sceHeatWaveSourceLoadMemory(), sceHeatWaveSourceSetType()

Document serial number: 000010343062

000010343062

sceHeatWaveSoundGetInfo

Gets information about a sound.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveSoundGetInfo(
    SceHwSound hSoundHandle,
    SceHwSoundInfo *pOutInfo
);
```

Arguments

[in] *hSoundHandle*	The sound to query.
[out] *pOutInfo*	Receives the information about a sound.

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Gets information about a sound. This consists of the number of sound effects that reference the sound and the number of sound sources that belong to the sound. The information can be used in conjunction with sceHeatWaveSoundGetSfx() and sceHeatWaveSoundGetSource() to loop through a sound's sound effects and sound sources.

See Also

sceHeatWaveSoundGetSfx(), sceHeatWaveSoundGetSource(), sceHeatWaveSoundGetSfxCount, sceHeatWaveSoundGetSourceCount()

000010343062

Document serial number: 000010343062

sceHeatWaveSoundGetSfx

Gets an existing sound effect, which references a particular sound, by index.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveSoundGetSfx(
    SceHwSound hSoundHandle,
    int32_t nIndex,
    SceHwSfx *pOutSfxHandle
);
```

Arguments

[in] *hSoundHandle*	The sound that is referenced by the sound effect.
[in] *nIndex*	The index of the sound effect to retrieve.
[out] *pOutSfxHandle*	Receives the sound effect.

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Gets an existing sound effect, which references a particular sound, by index. The sound effects referencing a sound are held in a zero-based array.

See Also

sceHeatWaveSoundGetSfxCount(), sceHeatWaveSoundCreateSfx(), sceHeatWaveReleaseSfx()

000010343062

Document serial number: 000010343062

sceHeatWaveSoundGetSfxCount

Gets the number of sound effects that reference a particular sound.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveSoundGetSfxCount(
    SceHwSound hSoundHandle,
    int32_t *pOutSfxCount
);
```

Arguments

[in] *hSoundHandle*	The sound to query.
[out] *pOutSfxCount*	Receives the number of sound effects that reference the sound if the operation was successful.

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Gets the number of sound effects that reference a particular sound. This function can be used in conjunction with sceHeatWaveSoundGetSfx() to loop through all the sound effects that reference a sound.

See Also

sceHeatWaveSoundGetSfx(), sceHeatWaveSoundCreateSfx(), sceHeatWaveReleaseSfx()

000010343062

Document serial number: 000010343062

sceHeatWaveSoundGetSource

Gets an existing sound source belonging to a sound by index.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveSoundGetSource(
    SceHwSound hSoundHandle,
    int32_t nIndex,
    SceHwSource *pOutSourceHandle
);
```

Arguments

[in] *hSoundHandle*	The sound to get the sound source from.
[in] *nIndex*	The index of the sound source to retrieve.
[out] *pOutSourceHandle*	Receives the sound source if the operation was successful.

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Gets an existing sound source belonging to a sound by index. The sound sources belonging a sound are held in a zero-based array.

See Also

sceHeatWaveSoundGetSourceCount(), sceHeatWaveSoundCreateSource(), sceHeatWaveReleaseSource()

000010343062

Document serial number: 000010343062

sceHeatWaveSoundGetSourceCount

Gets the number of sound sources that belong to a particular sound.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveSoundGetSourceCount(
    SceHwSound hSoundHandle,
    int32_t *pOutSourceCount
);
```

Arguments

[in] *hSoundHandle*	The sound to query.
[out] *pOutSourceCount*	Receives the number of sound sources that belong to the sound if the operation was successful.

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Gets the number of sound sources that belong to a particular sound. This function can be used in conjunction with sceHeatWaveSoundGetSource() to loop through all the sound sources that belong to a sound.

See Also

sceHeatWaveSoundGetSource(), sceHeatWaveSoundCreateSource(), sceHeatWaveReleaseSource()

000010343062

Document serial number: 000010343062

sceHeatWaveSoundSetLoopCount

Set a sound to loop on completion of all child sources.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveSoundSetLoopCount(
    SceHwSound hSoundHandle,
    int32_t nLoopCount
);
```

Arguments

[in] *hSoundHandle*	The sound to set.
[in] *nLoopCount*	Number of times to repeat. A value of 1 indicates that the sound should be played once and then repeated once. A value of 0 indicates that the sound should not be looped at all.

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Set a sound to loop on completion of all child sources. A sound effect referencing this sound will repeat when all sources have completed playback.

Notes

If *nLoopCount* is set to knLoopInfinite, the audio will continue to be output until sceHeatWaveSfxStop() is called.

000010343062

Document serial number: 000010343062

sceHeatWaveSourceGetInfo

Gets the current information about a particular sound source.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveSourceGetInfo(
    SceHwSource hSourceHandle,
    SceHwSourceInfo *pOutInfo
);
```

Arguments

[in] *hSourceHandle*	The sound source to query.
[out] *pOutInfo*	Receives the current information about the sound source if the operation was successful.

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Gets the current information about a particular sound source. This consists of the parent sound the sound source belongs to, the number of times the sound source is set to loop on playback and whether the sound is 2D or 3D.

See Also

sceHeatWaveSoundCreateSource(), sceHeatWaveSoundGetSource(), sceHeatWaveSourceSetLoopCount(), sceHeatWaveSourceSetPan(), sceHeatWaveSourceSetType(), sceHeatWaveSourceSetCone(), sceHeatWaveSourceSetRolloffDistances()

000010343062

Document serial number: 000010343062

sceHeatWaveSourceLoadFile

Sets the audio file for a sound source to use.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveSourceLoadFile(
    SceHwSource hSourceHandle,
    const char *strFilePath
);
```

Arguments

[in] *hSourceHandle*	The audio source that will own the audio file.
[in] *strFilePath*	The path to the audio file.

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Sets the audio file for a sound source to use. Once this function has been called, the audio from the supplied file will be used when the parent sound is required to play.

Notes

MP3 and WAV file format (16-bit PCM only) are supported.

000010343062

Document serial number: 000010343062

sceHeatWaveSourceLoadMemory

Sets a sound source to use some audio data loaded in memory.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveSourceLoadMemory(
    SceHwSource hSourceHandle,
    SceHwMemoryFileInfo *pMemoryFile
);
```

Arguments

[in] *hSourceHandle*	The audio source that will own the audio data.
[in] *pMemoryFile*	The location in memory of the audio data.

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Sets a sound source to use some audio data loaded in memory. Once this function has been called, the supplied audio data will be used when the parent sound is required to play. The audio data should come from a previously loaded audio file.

Notes

MP3 and WAV file format (16-bit PCM only) is supported.

000010343062

Document serial number: 000010343062

©SIE Inc

sceHeatWaveSourceLoadRuntimeStream

Set the callback used to generate audio data for instances of the `SceHwSource`.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveSourceLoadRuntimeStream(
    SceHwSource hSourceHandle,
    SceHwSourceRuntimeStreamParam *pParam,
    SceHwSourceRuntimeStreamCallbackFunction pFunc,
    void *pUserData = NULL
);
```

Arguments

[in] *hSourceHandle*	The audio source that will own the audio data.
[in] *pParam*	Information about the stream.
[in] *pFunc*	The callback function used to generate audio data.
[in] *pUserData*	User specified data passed to the callback function (optional).

Return Values

See `SCE_HEATWAVE_API_OK` and other return codes.

Description

Set the callback used to generate audio data for instances of the `SceHwSource`. Once this function has been called the supplied callback will be used to generate audio data for each instance of the `SceHwSource`.

Notes

16 bit PCM only is supported.

See Also

SceHwSourceRuntimeStreamCallbackFunction

000010343062

Document serial number: 000010343062

sceHeatWaveSourceSetCone

Sets the 3D cone for a sound source.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveSourceSetCone(
    SceHwSource hSourceHandle,
    float32_t fInside,
    float32_t fOutside,
    float32_t fOutsideGain,
    float32_t fOutsideFreq
);
```

Arguments

[in] *hSourceHandle*	The sound source to update the 3D cone for.
[in] *fInside*	The inside angle of the cone. This should be between `kfSourceConeAngleMin` and `kfSourceConeAngleMax` degrees.
[in] *fOutside*	The outside angle of the cone. This should be between `kfSourceConeAngleMin` and `kfSourceConeAngleMax` degrees.
[in] *fOutsideGain*	The volume of the sound source at angles greater than *fOutside*. This should be between `kfSourceConeGainMin` and `kfSourceConeGainMax`.
[in] *fOutsideFreq*	The highest frequency cutoff point for the sound source at angles greater than *fOutside*. This should be between `kfSourceConeFreqMin` and `kfSourceConeFreqMax`.

Return Values

See `SCE_HEATWAVE_API_OK` and other return codes. A value of `SCE_HEATWAVE_API_ERROR_OPERATION` will be returned if an attempt is made to set a sound cone for a 2D sound source.

Description

Sets the 3D cone for a sound source. The volume and high frequency content of the sound source audio varies according to the angle between the direction the sound source is facing and the direction the listener is facing. If this angle is greater than *fOutside*, the sound will not be further attenuated or experience more high frequency filtering even if the angle increases. In such a case, the volume will remain at *fOutsideGain* and the highest frequency cutoff point will remain at *fOutsideFreq*. If the angle is less than *fInside*, the sound will not get louder or contain more high frequencies even if the angle decreases. The amount of volume reduction and high frequency filtering linearly interpolates between *fInside* and *fOutside*.

The values set using this function work with the values set by `sceHeatWaveSourceSetRolloffDistances()` to work out the volume of a sound source when it is played as part of a sound effect. This means the volume of the sound source in a sound effect will result from three things. The first is the listener's position inside/outside of the sound effect's sound cone, which depends on values set by this function and the sound effect's direction. The second is the distance of the listener from the sound effect's point of emission, which depends on values set by `sceHeatWaveSourceSetRolloffDistances()` and the sound effect's position. The third is the volume setting of the sound effect itself.

Document serial number: 000010343062

000010343062

See Also

sceHeatWaveSourceSetRolloffDistances()

000010343062

Document serial number: 000010343062

sceHeatWaveSourceSetLoopCount

Sets a sound source to loop a number of times.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveSourceSetLoopCount(
    SceHwSource hSourceHandle,
    int32_t nLoopCount
);
```

Arguments

[in] *hSourceHandle*	The sound source to loop.
[in] *nLoopCount*	The number of times the sound source should repeat. A value of 1 indicates that the sound should be played once and then repeated once. A value of 0 indicates that the sound should not be looped at all.

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Sets a sound source to loop a number of times. Each time the parent sound is played (via a sound effect), the sound source will be repeated this many additional times.

Notes

If *nLoopCount* is set to knLoopInfinite, the audio will continue to be output until sceHeatWaveSfxStop() is called.

000010343062

Document serial number: 000010343062

sceHeatWaveSourceSetPan

Sets the pan position for a sound source.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveSourceSetPan(
    SceHwSource hSourceHandle,
    float32_t fPan
);
```

Arguments

[in] *hSourceHandle*	The sound source to apply the pan position to.
[in] *fPan*	The new pan position for the sound source, which should be between `kfSourcePanLeft` and `kfSourcePanRight`.

Return Values

See `SCE_HEATWAVE_API_OK` and other return codes. A value of `SCE_HEATWAVE_API_ERROR_OPERATION` will be returned if an attempt is made to set the pan position for a 3D sound source.

Description

Sets the pan position for a sound source. This function sets the position of a sound in a 2D stereo space.

See Also

`sceHeatWaveSourceSetType()`

000010343062

Document serial number: 000010343062

©SIE Inc

sceHeatWaveSourceSetPitchMultiplier

Sets the playback pitch for a sound source.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveSourceSetPitchMultiplier(
    SceHwSource hSourceHandle,
    float32_t fPitchMultiplier
);
```

Arguments

[in] *hSourceHandle*	The sound source to set the pitch multiplier for.
[in] *fPitchMultiplier*	The new pitch multiplier for the sound source. This must be a value between `kfSourcePitchMin` and `kfSourcePitchMax`.

Return Values

See `SCE_HEATWAVE_API_OK` and other return codes.

Description

Sets the playback pitch for a sound source.

000010343062

Document serial number: 000010343062

sceHeatWaveSourceSetRolloffDistances

Sets the distance over which the volume of a particular sound source will change (roll off) in relation to the distance of the listener from the sound.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveSourceSetRolloffDistances(
    SceHwSource hSourceHandle,
    float32_t fMinDistance,
    float32_t fMaxDistance
);
```

Arguments

[in] *hSourceHandle*	The sound source to update the roll-off distance for.
[in] *fMinDistance*	The distance below which the volume will no longer be increased.
[in] *fMaxDistance*	The distance above which the volume will no longer be attenuated.

Return Values

See SCE_HEATWAVE_API_OK and other return codes. A value of SCE_HEATWAVE_API_ERROR_OPERATION will be returned if an attempt is made to set a roll-off distance for a 2D sound source.

Description

Sets the distance over which the volume of a particular sound source will change (roll off) in relation to the distance of the listener from the sound. If the sound emission point is further than *fMaxDistance* from the listener, the sound will not be further attenuated even if it moves farther away. If the sound emission point is nearer than *fMinDistance* to the listener, the sound will not get louder even if it gets closer. Sound roll-off occurs between these two points and follows an inverse distance model where gain reduces exponentially as distance increases.

The values set using this function work with the values set by sceHeatWaveSourceSetCone() to work out the volume of a sound source when it is played as part of a sound effect. This means the volume of the sound source in a sound effect will result from three things. The first is the listener's position inside/outside of the sound effect's sound cone, which depends on values set by sceHeatWaveSourceSetCone() and the sound effect's direction. The second is the distance of the listener from the sound effect's point of emission, which depends on values set by this function and the sound effect's position. The third is the volume setting of the sound effect itself.

See Also

sceHeatWaveSourceSetCone()

Document serial number: 000010343062

000010343062

sceHeatWaveSourceSetType

Sets the sound source to be a 3D or 2D emitter.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveSourceSetType(
    SceHwSource hSourceHandle,
    SceHwSourceType eType
);
```

Arguments

[in] *hSourceHandle*	The sound source to set.
[in] *eType*	The sound source emitter type.

Return Values

See SCE_HEATWAVE_API_OK and other return codes.

Description

Sets the sound source to be a 3D or 2D emitter.

See Also

sceHeatWaveSourceGetInfo(), sceHeatWaveSourceSetLoopCount(), sceHeatWaveSourceSetPan(), sceHeatWaveSourceSetType(), sceHeatWaveSourceSetCone(), sceHeatWaveSourceSetRolloffDistances()

0000103430 62

Document serial number: 000010343062

©SIE Inc

sceHeatWaveSourceSetVolume

Sets the playback volume (gain) of a sound source.

Definition

```
#include <heatwave_api.h>
SCE_HEATWAVE_FUNC SceHwError sceHeatWaveSourceSetVolume(
    SceHwSource hSourceHandle,
    float32_t fVolume
);
```

Arguments

[in] *hSourceHandle*	The sound source to set the volume for.
[in] *fVolume*	The new volume for the sound source. This must be a value between `kfSourceVolumeMin` and `kfSourceVolumeMax`.

Return Values

See `SCE_HEATWAVE_API_OK` and other return codes.

Description

Sets the playback volume (gain) of a sound source.

0000103430 62

Document serial number: 000010343062

Constants

000010343062

Document serial number: 000010343062

©SIE Inc

Constant Summary

Constant	Value	Description
kfSfxPitchMax	2.0f	Sets the maximum SceHwSfx pitch multiplier.
kfSfxPitchMin	0.0f	Sets the minimum SceHwSfx pitch multiplier.
kfSfxVolumeMax	1.0f	Sets the maximum SceHwSfx volume multiplier.
kfSfxVolumeMin	0.0f	Sets the minimum SceHwSfx volume multiplier.
kfSourceConeAngleMax	360.0f	Sets the maximum angle an emitter cone can have.
kfSourceConeAngleMin	0.0f	Sets the minimum angle an emitter cone can have.
kfSourceConeFreqMax	1.0f	Sets the maximum frequency an emitter cone can have.
kfSourceConeFreqMin	0.0f	Sets the minimum frequency an emitter cone can have.
kfSourceConeGainMax	1.0f	Sets the maximum gain an emitter cone can have.
kfSourceConeGainMin	0.0f	Sets the minimum gain an emitter cone can have.
kfSourcePanLeft	-1.0f	Sets the maximum left pan a SceHwSource can have.
kfSourcePanRight	1.0f	Sets the maximum right pan a SceHwSource can have.
kfSourcePitchMax	2.0f	Sets the maximum SceHwSource pitch multiplier.
kfSourcePitchMin	0.0f	Sets the minimum SceHwSource pitch multiplier.
kfSourceVolumeMax	1.0f	Sets the maximum SceHwSource volume multiplier.
kfSourceVolumeMin	0.0f	Sets the minimum SceHwSource volume multiplier.
kfSpeedOfSound	343.3f	Defines the speed of sound in meters per second.
knLoopInfinite	-1	Sets a SceHwSource to loop until sceHeatWaveSfxStop() is called.
knSfxAudio3dPriorityMax	0	The highest priority an Audio3d SceHwSfx can have.
knSfxAudio3dPriorityMin	INT_MAX	The lowest priority a SceHwSfx Audio3d object can have.
kuSceHwInvalidHandle	ULLONG_MAX	Constant for a handle that is not valid.

000010343062

Document serial number: 000010343062

Defines

000010343062

Document serial number: 000010343062

Define Summary

Define	Value	Description
SCE_HEATWAVE_API_VERSION	0x1043D87E	The version number in hex format (for example 0x10132AD7). The format of the version number is MmmXXXXX where M is the major version, mm is the minor version and XXXXX is the revision.
SCE_HEATWAVE_FUNC	N/A	Marker for HeatWave API functions. Used to specify the API functions as dynamic linkage.
SCE_HEATWAVE_API_NO_CALLBACK	(NULL)	Do not execute a user callback.
SCE_HEATWAVE_API_OK	0	The operation was successful.
SCE_HEATWAVE_API_ERROR	-2147483647	(0x80000001) The operation was not successful.
SCE_HEATWAVE_API_ERROR_OUT_OF_ASSETS	-2147483646	(0x80000002) The operation failed because there was insufficient memory available.
SCE_HEATWAVE_API_ERROR_PARAM	-2147483645	(0x80000003) The operation failed because a parameter was invalid or missing.
SCE_HEATWAVE_API_ERROR_HANDLE	-2147483644	(0x80000004) The operation failed because the supplied handle was not recognized as a HeatWave handle.
SCE_HEATWAVE_API_ERROR_FILE	-2147483643	(0x80000005) The operation failed because a file was missing.
SCE_HEATWAVE_API_ERROR_NO_HARDWARE_CONTEXT	-2147483642	(0x80000006) The operation failed because there is no hardware context.
SCE_HEATWAVE_API_ERROR_INVALID_OPERATION	-2147483641	(0x80000007) The operation failed because it is not supported.
SCE_HEATWAVE_API_ERROR_NO_OUTPUT_DEVICE	-2147483640	(0x80000008) The operation failed because there is no audio output device.
SCE_HEATWAVE_API_ERROR_UNSUPPORTED_FORMAT	-2147483639	(0x80000009) The operation failed because a file had an invalid format.
SCE_HEATWAVE_API_OUTPUT_MAIN	(0)	Main output.
SCE_HEATWAVE_API_OUTPUT_AUX	(1)	Auxiliary output.
SCE_HEATWAVE_API_OUTPUT_PAD_1	(2)	Pad speaker of Player 1.
SCE_HEATWAVE_API_OUTPUT_PAD_2	(3)	Pad speaker of Player 2.
SCE_HEATWAVE_API_OUTPUT_PAD_3	(4)	Pad speaker of Player 3.
SCE_HEATWAVE_API_OUTPUT_PAD_4	(5)	Pad speaker of Player 4.
SCE_HEATWAVE_API_OUTPUT_AUDIO3D	(6)	Audio3d object.
SCE_HEATWAVE_API_OUTPUT_MAX	(7)	Number of supported outputs.

Document serial number: 000010343062

000010343062